INSTRUCTIONS

AF3060 INVERTER DRIVES

INVERTER ASSEMBLY - 331X324BAG01, G04
G02, G05
G03, G06

OPERATION - TROUBLE SHOOTING - REPAIR GEK 62503

DIAGRAMS

INVERTER CONV ASM	ELEM	36C764816AB
	CONN	36D865531AA (G01, G04)
		36D865530AB (G02, G05)
		36D865520AB (G03, G06)
PHASE MODULE	ELEM	36C764814AA
	CONN	36C764815AA
RECTIFIER MODULE ASM	ELEM	36C764812AA
	CONN	36C764813AA
CURRENT FEEDBACK	ELEM	36C764112AB
PULSE TRANSFORMER	ELEM	36C762856AB
AUXILIARY SUPPRESSION	ELEM	36C762897BA

These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to the General Electric Company.

DUPLICATE COPY 6/27/79

GENERAL ELECTRIC

GENERAL ELECTRIC

36C764816AB
CONT ON SHEET 2 SH NO 1

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE ELEMENTARY DIAGRAM
AF-3061 INVERTER ASM
FIRST MADE FOR STANDARD LINE
331X324BAG01-G06

FAN ASM (SEE NOTE 8)
B MOT — FUI — BMTB — (BM1) (BM2)

THSW

PTB (PI) (PIR)
5TB (10) (11) THI TH2
(SEE NOTE 9) SEE NOTE 7
RECT (RECTIFIER MODULE)
ELEM DIAG 36C764812AA
CONN DIAG 36C764813AA
CAT NO 331X322AA G02
230VAC INPUT (L1) (L2) (L3) L1 L2 L3
PI (+) P2 (-)
(P2) P2
LI (SEE NOTE 2) (P3) P3
CTF3

6TB (PI) (AXI)
(29) (30)
(P3) (AX3) 6TB

ARP
R10 100K ACP1 400 MF
R11 100K ACP2 400 MF

(29) AUXILIARY SUPPRESSION ASSEMBLY ASA (30)
(AM) (BM) (CM) 3TB
(AN) (BN) (CN) 4TB

ACN1 400 MF R12 100K
ACN2 400 MF R13 100K
ARN

CPTB (PI) CPTB' (P3)
C1 C1
R1 Q 1K 300W
CTF

PMA (PHASE MODULE A)
ELEM DIAG 36C764814AA
CONN DIAG 36C764815AA
CAT NO. 331X323AAG02,3
PMB (SAME AS PHASE A)
PMC (SAME AS PHASE A)
PI P3 N M
APL BPL CPL

CTCA CCA AN AL LCA
CTCB CCB BN BL LCB
CTCC CCC CN CL LCC
CTA CTB CTC
PTB (TI) TI
T2 (T2)
T3 (T3)

5TB (1) (2) (3) (4) (5) (6)
W R W R W R

CURRENT FEEDBACK PANEL ASSEMBLY
ITB (PIF) ITB (QF) ITB (RFN) ITB (RFP) ITB (F3N)
R3 1K R2 1K RF SH 2 R4 47K R5
ITB (PIL) ITB (QL) ITB (PID) ITB (QD) ITB (RFD)
D2 D1 D3
7 8 16 15

CURRENT FEEDBACK CARDS
ACF
BCF
CCF
K1 K2 K3 K4
J1 J2 J3 J4
17 18 19 20
21 22 23 24
25 26 27 28
SEE NOTE 6
.08MF 10K
SEE NOTE 16

THI TH2 (THI) (TH2)
1PL 2PL

SEE SHEET 2 FOR NOTES AND COMPONENT VALUES

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department

REVISIONS	
1	J.R. Baker JAN 15, 1974

PRINTS TO	
5B(3)23	JA(CD)
5D(BK)	IT(BW)
5(QC)B	5AE(BW)
5QC(BK)5	5L(BW)5
5(TS)50	
5(S)	

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

MADE BY V. Babo MAR. 30, 1973
for ltr D.J. Lantos Aug. 3, 1973
APPROVALS

SPEED VARIATOR DIV OR DEPT ERIE, PA LOCATION
36C764816AB CONT ON SHEET 2 SH NO 1

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE NOTES
AF-3061 INVERTER ASM
FIRST MADE FOR STANDARD LINE
33|X324BAG01-G06

1. THE ASSEMBLY CATALOG NUMBER GROUP DISTINCTION, MODULE CATALOG NUMBERS AND SOME COMPONENT VALUES ARE DEPENDENT UPON DVA RATING. TABLE I LISTS VARIABLE COMPONENT VALUES.

2. THE L1 CHOKE IS MOUNTED REMOTE TO THE REMAINDER OF THE INVERTER ASSEMBLY, ITS VALUE IS DEPENDENT UPON KVA RATING AND IS LISTED IN TABLE I.

3. ELEMENTARY WIRING SYMBOLS



W / R — INDICATES TWISTED WIRES, COLOR INDICATED BY LETTER

1 / 2 / 3 — INDICATES CABLE ASSEMBLY, NUMBERS ARE POINTS IN CONNECTOR PLUG

DOT INDICATES POSITIVE TERMINAL WHEN CURRENT FLOWS IN THE DIRECTION OF THE ARROW.

4. REFER TO THE CONNECTION DIAGRAM FOR A PHYSICAL REPRESENTATION OF THIS ASSEMBLY.

5. ON PRINTED CIRCUIT CARDS USED IN THIS ASSEMBLY, THE LETTERS "AA" AFTER THE BASIC CATALOG NUMBER INDICATES ORIGINAL DESIGN. SUBSEQUENT DESIGNS WITH THE SAME BASIC NUMBER AND GROUP NUMBER WITH THE SECOND LETTER CHANGED, SUCH AS "AB", "AC", "AD", ETC. ARE DIRECTLY INTERCHANGEABLE AND MAY BE SUPPLIED IN PLACE OF THE "AA" CARD. AN EARLIER DESIGN CANNOT REPLACE A LATER DESIGN, E.G. AN "AA" CANNOT REPLACE AN "AB"

6. THE NOTED RESISTOR VALUES ON THE CURRENT FEEDBACK CARD ARE DEPENDENT UPON KVA RATING. THE PARTICULAR CARD AND JUMPER CONNECTION IS FOUND IN TABLE I. THE CIRCUIT SHOWN IN DASHED LINES IS A REPRESENTATION OF THE ACTUAL CIRCUITRY AND IS IDENTICAL FOR ACF, BCF AND CCF.

7. RECTIFIER ASSEMBLY AND ASSOCIATED WIRE SHOWN IN DASHED LINES ARE OMITTED FOR G04-G06.

8. FAN ASSEMBLY AND ASSOCIATED WIRES SHOWN IN DASHED LINES ARE SUPPLIED ON G01 AND G04 ONLY WHEN SPECIFIED. SEE SYSTEM DIAGRAMS FOR ACTUAL CIRCUITRY.

9. PIR JUMPER SHOWN AS A DASHED LINE IS REQUIRED FOR G01 AND G02.

TABLE I

KVA RATING	L1	C1	C1 & C1	CCA, CCB, CCC	LCA, LCB, LCC	RF RF3	ACF BCF CCF	CURRENT FEEDBACK JUMPER KVA	CURRENT FEEDBACK JUMPER CONNECTION	ARP ARN
10	3MH	5700μF	2850μF	5μF	28μH	120Ω	193X 199BAG01	5	XA	2500Ω
								7.5	XB	
								10	XC	
25	2MH	11400μF	5700μF	10μF	12μH	56.2Ω	193X 199BAG02	15	XA	1000Ω
								20	XB	
								25	XC	
50	.8MH	19000μF	9500μF	20μF	4μH	33 2Ω	193X 199BAG03	30	XA	500Ω
								40	XB	
								50	XC	

FOLD

DRAWING NO
CONT ON SHT
SH NO
REV NO
B
2

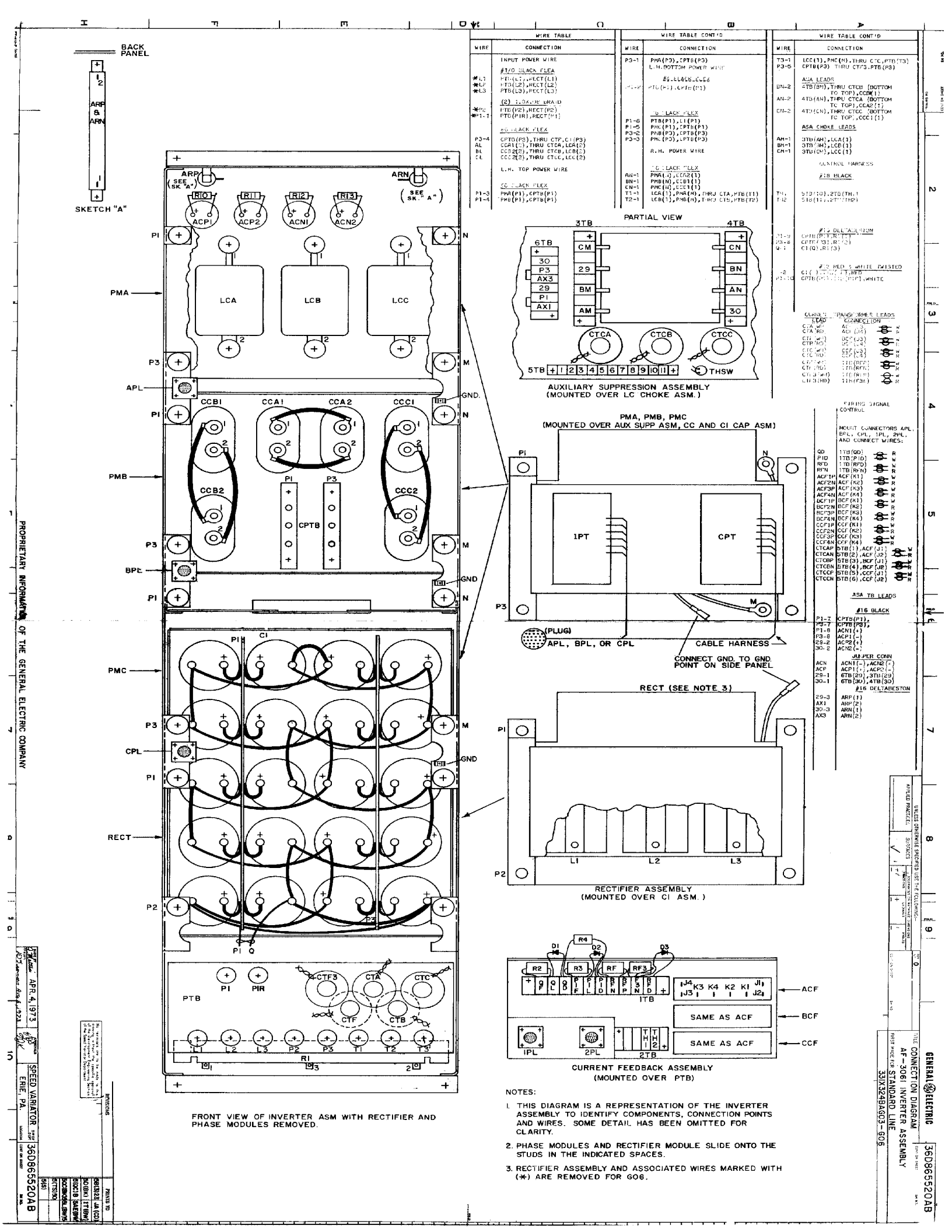
No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department

REVISIONS
2 DW Hoover 3/14
REDRAWN GEN CH

PRINTS TO

5B(8)M	5J(BW)
5R(2)BW	TS(50)
5L(2)BW	
AW(BW)	
5D(CD)	
JA(CD)	

MADE BY BaBo MAR 30, 1973
APPROVALS
REV ISSUED D.S. Hoover Aug 11, 1973
SPEED VARIATOR DIV OR DEPT
ERIE, PA, LOCATION
36C764816AB
CONT ON SHEET FL SH NO 2



WIRE	CONNECTION
	INPUT POWER WIRE
	#1/0 BLACK FLEX
*L1	PTB(L1),,RECT(L1)
*L2	PTB(L2),RECT(L2)
*L3	PTB(L3),RECT(L3)
	(2) #4/0 BRAID
*P2	PTB(P2),RECT(P2)
*P1-1	PTB(P1R),RECT(P1)
	#6 BLACK FLEX
P3-4	CPTB(P3),THRU CTF,CI(P3)
AL	CCA1(1),THRU CTCA,LCA(2)
BL	CCB2(2),THRU CTCB,LCB(2)
CL	CCC2(2),THRU CTCC,LCC(2)
	L.H. TOP POWER WIRE
	#2 BLACK FLEX
P1-3	PMA(P1),,CPTB(P1)
P1-4	PMB(P1),,CPTB(P1)

WIRE	CONNECTION
P3-1	PMA(P3),,CPTB(P3)
	L.H. BOTTOM POWER WIRE
	#6 BLACK FLEX
P1-2	PTB(P1),,CPTB(P1)
	#8 BLACK FLEX
P1-6	PTB(P1),,LI(P1)
P1-5	PMC(P1),,CPTB(P1)
P3-2	PMB(P3),,CPTB(P3)
P3-3	PMC(P3),,CPTB(P3)
	R.H. POWER WIRE
	#8 BLACK FLEX
AN-1	PMA(N),,CCA2(1)
BN-1	PMB(N),,CCB1(1)
CN-1	PMC(N),,CCC1(1)
T1-1	LCA(1),,PMA(M),,THRU CTA,,PTB(T1)
T2-1	LCB(1),,PMB(M),,THRU CTB,,PTB(T2)

WIRE	CONNECTION
T3-1	LCC(1),,PMC(M),,THRU CTC,,PTB(T3)
P3-5	CPTB(P3) THRU CTF3,,PTB(P3)
	#4 LEADS
BN-2	4TB(BN),,THRU CTCB (BOTTOM TO TOP),,CCB(1)
AN-2	4TB(AN),,THRU CTCA (BOTTOM TO TOP),,CCA2(1)
CN-2	4TB(CN),,THRU CTCC (BOTTOM TO TOP),,CCC1(1)
	#8 CHOKE LEADS
AM-1	3TB(AM),,LCA(1)
BM-1	3TB(BM),,LCB(1)
CM-1	3TB(CM),,LCC(1)
	CONTROL HARNESS
	#18 BLACK
TH1	5TB(10),,2TB(TH1)
TH2	5TB(11),,2TB(TH2)

P1-9	CPTB(P1T),,RT(1)
P3-6	CPTB(P3T),,RT(3)
Q-1	CI(Q),,RT(3)
-2	CI(),,RT(RFD)
P2-10	CPTB(),,RT(P2),WHITE

CURRENT TRANSFORMER LEADS

LEAD	CONNECTION
CTA (AP)	ACI (J4)
CTA (AN)	ACI (J3)
CTB (BP)	BCF (J3)
CTB (BN)	BCF (J4)
CTC (CP)	CCF (J3)
CTC (CN)	CCF (J4)
CT1 (AP)	1ID (RFD)
CT1 (AN)	1ID (RFN)
CT2 (BP)	1ID (P3T)
CT3 (CN)	1ID (P3T)

QD	1TB(QD)
P1D	1TB(P1D)
RFD	1TB(RFD)
RFN	1TB(RFN)
ACF1P	ACF(K1)
ACF2N	ACF(K2)
ACF3P	ACF(K3)
ACF4N	ACF(K4)
BCF1P	BCF(K1)
BCF2N	BCF(K2)
BCF3P	BCF(K3)
BCF4N	BCF(K4)
CCF1P	CCF(K1)
CCF2N	CCF(K2)
CCF3P	CCF(K3)
CCF4N	CCF(K4)
CTCAP	5TB(1),,ACF(J1)
CTCAN	5TB(2),,ACF(J2)
CTCBP	5TB(3),,BCF(J1)
CTCBN	5TB(4),,BCF(J2)
CTCCP	5TB(5),,CCF(J1)
CTCCN	5TB(6),,CCF(J2)

ASA TB LEADS

#16 BLACK

P1-7	CPTB(P1),
P3-7	CPTB(P3),
P1-8	ACN1(+)
P3-8	ACP1(-)
29-2	ACP2(+)
30-2	ACN2(-)

JUMPER CONN

ACN	ACN1(-),,ACN2(-)
ACP	ACP1(+),,ACP2(+)
29-1	6TB(29),,3TB(29)
30-1	6TB(30),,4TB(30)

#16 DELTABESTON

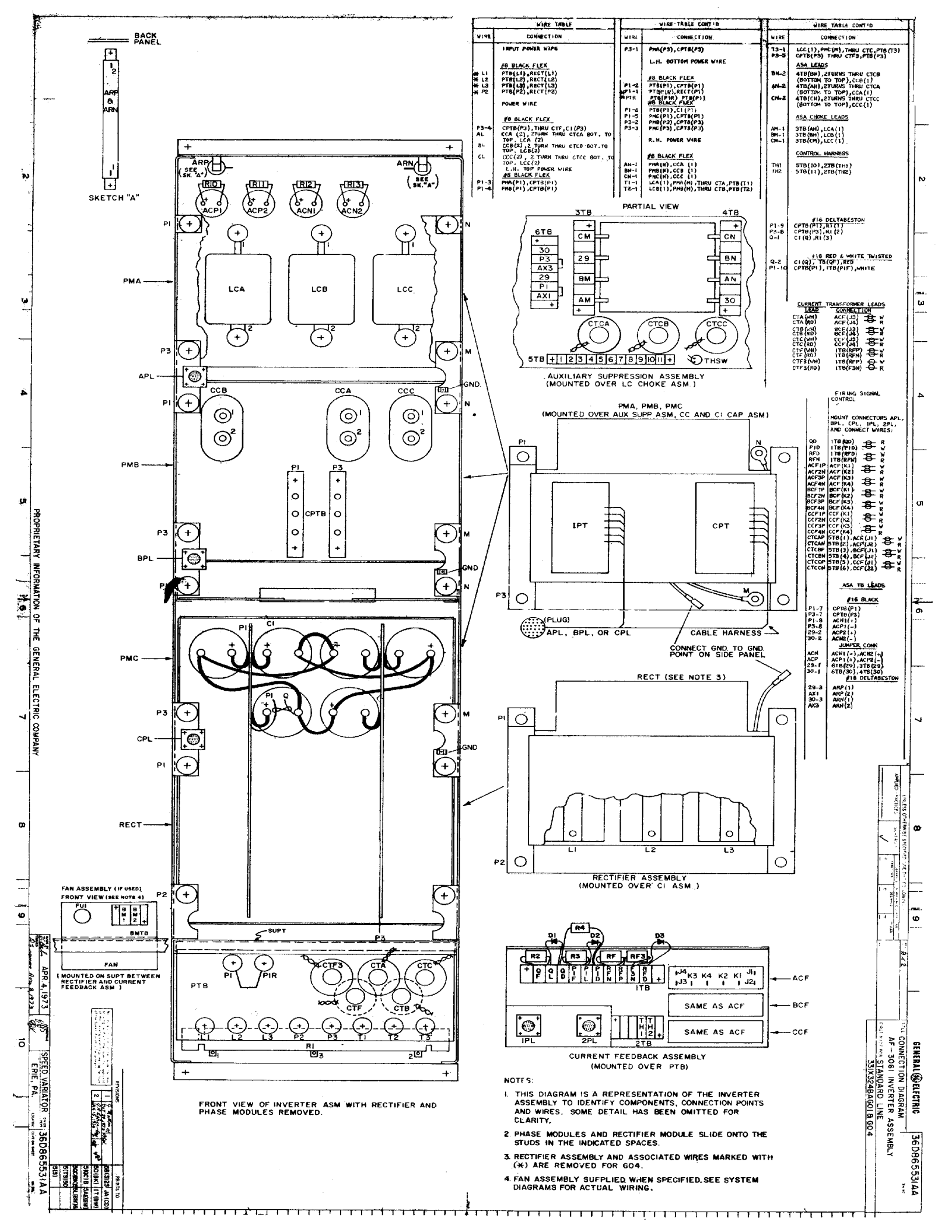
29-3	ARP(1)
AX1	ARP(2)
30-3	ARN(1)
AX3	ARN(2)

NOTES:

1. THIS DIAGRAM IS A REPRESENTATION OF THE INVERTER ASSEMBLY TO IDENTIFY COMPONENTS, CONNECTION POINTS AND WIRES. SOME DETAIL HAS BEEN OMITTED FOR CLARITY.
2. PHASE MODULES AND RECTIFIER MODULE SLIDE ONTO THE STUDS IN THE INDICATED SPACES.
3. RECTIFIER ASSEMBLY AND ASSOCIATED WIRES MARKED WITH (*) ARE REMOVED FOR 606.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING—
SURFACES
TOLERANCES
DECIMALS
ANGULAR
FRACTIONS
MATERIAL
FINISH
DO NOT SCALE DRAWING
APPROVALS
DRAWN
CHECKED
ENGR APPROVAL
MFG APPROVAL
COMMERCIAL
NEXT ASSY
APPLICATION
DO NOT REMOVE THIS BORDER



WIRE	CONNECTION
	INPUT POWER WIRE
	#8 BLACK FLEX
* L1	PTB(L1),RECT(L1)
* L2	PTB(L2),RECT(L2)
* L3	PTB(L3),RECT(L3)
* P2	PTB(P2),RECT(P2)
	POWER WIRE
	#8 BLACK FLEX
P3-4	CPTB(P3),THRU CTF,C1(P3)
AL	CCA (2), 2TURN THRU CTCA BOT. TO TOP, LCA (2)
BL	CCB(2), 2 TURN THRU CTCB BOT.TO TOP, LCB(2)
CL	CCC(2), 2 TURN THRU CTCC BOT. TO TOP, LCC(2)
	L.H. TOP POWER WIRE
	#8 BLACK FLEX
P1-3	PMA(P1),CPTB(P1)
P1-4	PMB(P1),CPTB(P1)

WIRE	CONNECTION
P3-1	PMA(P3),CPTB(P3)
	L.H. BOTTOM POWER WIRE
	#8 BLACK FLEX
P1-2	PTB(P1),CPTB(P1)
*P1-1	PTB(P1R),RECT(P1)
*P1R	PTB(P1R) P1B(P1)
	#8 BLACK FLEX
P1-6	PTB(P1),C1(P1)
P1-5	PMC(P1),CPTB(P1)
P3-2	PMB(P2),CPTB(P3)
P3-3	PMC(P3),CPTB(P3)
	R.H. POWER WIRE
	#8 BLACK FLEX
AN-1	PMA(N),CCA (1)
BN-1	PMB(N),CCB (1)
CN-1	PMC(N),CCC (1)
T1-1	LCA(1),PMA(M),THRU CTA,PTB(T1)
T2-1	LCB(1),PMB(M),THRU CTB,PTB(T2)

WIRE	CONNECTION
T3-1	LCC(1),PMC(M),THRU CTC,PTB(T3)
P3-5	CPTB(P3) THRU CTF3,PTB(P3)
	ASA LEADS
BN-2	4TB(BM),2TURNS THRU CTCB (BOTTOM TO TOP),CCB(1)
AN-2	4TB(AM),2TURNS THRU CTCA (BOTTOM TO TOP),CCA(1)
CN-2	4TB(CM),2TURNS THRU CTCC (BOTTOM TO TOP),CCC(1)
	ASA CHOKE LEADS
AM-1	3TB(AM),LCA(1)
BM-1	3TB(BM),LCB(1)
CM-1	3TB(CM),LCC(1)
	CONTROL HARNESS
TH1	5TB(10),2TB(TH1)
TH2	5TB(11),2TB(TH2)
	#16 DELTABESTON
P1-9	CPTB(P1),RT(1)
P3-8	CPTB(P3),RI(2)
Q-1	CI(Q),RI(3)
	#18 RED & WHITE TWISTED
Q-2	CI(Q), TB(QF),RED
P1-10	CPTB(P1),ITB(P1F),WHITE

CURRENT TRANSFORMER LEADS	
LEAD	CONNECTION
CTA (WH)	ACF (J3)
CTA (RD)	ACF (J4)
CTB (WH)	BCF (J3)
CTB (RD)	BCF (J4)
CTC (WH)	CCF (J3)
CTC (RD)	CCF (J4)
CTF (WH)	ITB (RFP)
CTF (RD)	ITB (RFN)
CTF3 (WH)	ITB (RFP)
CTF3 (RD)	ITB (F3N)

FIRING SIGNAL CONTROL	
MOUNT CONNECTORS APL, BPL, CPL, 1PL, 2PL, AND CONNECT WIRES:	
QD	1TB(QD)
PJD	1TB(P1D)
RFD	1TB(RFD)
RFN	1TB(RFN)
ACF1P	ACF(K1)
ACF2N	ACF(K2)
ACF3P	ACF(K3)
ACF4N	ACF(K4)
BCF1P	BCF(K1)
BCF2N	BCF(K2)
BCF3P	BCF(K3)
BCF4N	BCF(K4)
CCF1P	CCF(K1)
CCF2N	CCF(K2)
CCF3P	CCF(K3)
CCF4N	CCF(K4)
CTCAP	5TB(1),ACF(J1)
CTCAN	5TB(2),ACF(J2)
CTCBP	5TB(3),BCF(J1)
CTCBN	5TB(4),BCF(J2)
CTCCP	5TB(5),CCF(J1)
CTCCN	5TB(6),CCF(J2)
	ASA TB LEADS
	#16 BLACK
P1-7	CPTB(P1)
P3-7	CPTB(P3)
P1-8	ACP1(+)
P3-8	ACP1(-)
29-2	ACP2(+)
30-2	ACN2(-)
	JUMPER CONN
ACN	ACN1(-),ACN2(+)
ACP	ACP1(+),ACP2(-)
29-1	6TB(29),3TB(29)
30-1	6TB(30),4TB(30)
	#16 DELTABESTON
29-3	ARP(1)
AX1	ARP(2)
30-3	ARN(1)
AX3	ARN(2)

NOTES:

1. THIS DIAGRAM IS A REPRESENTATION OF THE INVERTER ASSEMBLY TO IDENTIFY COMPONENTS, CONNECTION POINTS AND WIRES. SOME DETAIL HAS BEEN OMITTED FOR CLARITY.
2. PHASE MODULES AND RECTIFIER MODULE SLIDE ONTO THE STUDS IN THE INDICATED SPACES.
3. RECTIFIER ASSEMBLY AND ASSOCIATED WIRES MARKED WITH (*) ARE REMOVED FOR G04.
4. FAN ASSEMBLY SUPPLIED WHEN SPECIFIED. SEE SYSTEM DIAGRAMS FOR ACTUAL WIRING.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

APR 4, 1973

SPEED VARIATOR TYPE 36D865531A

ERIE, PA.

REVISIONS			
2			
1			

PRINTS TO

6181	6T39150	6D10761 G1 (11 WH)	6B10325 JA1 (10)
		6D6618 (ASSEMBLY DRAWING)	

GENERAL ELECTRIC

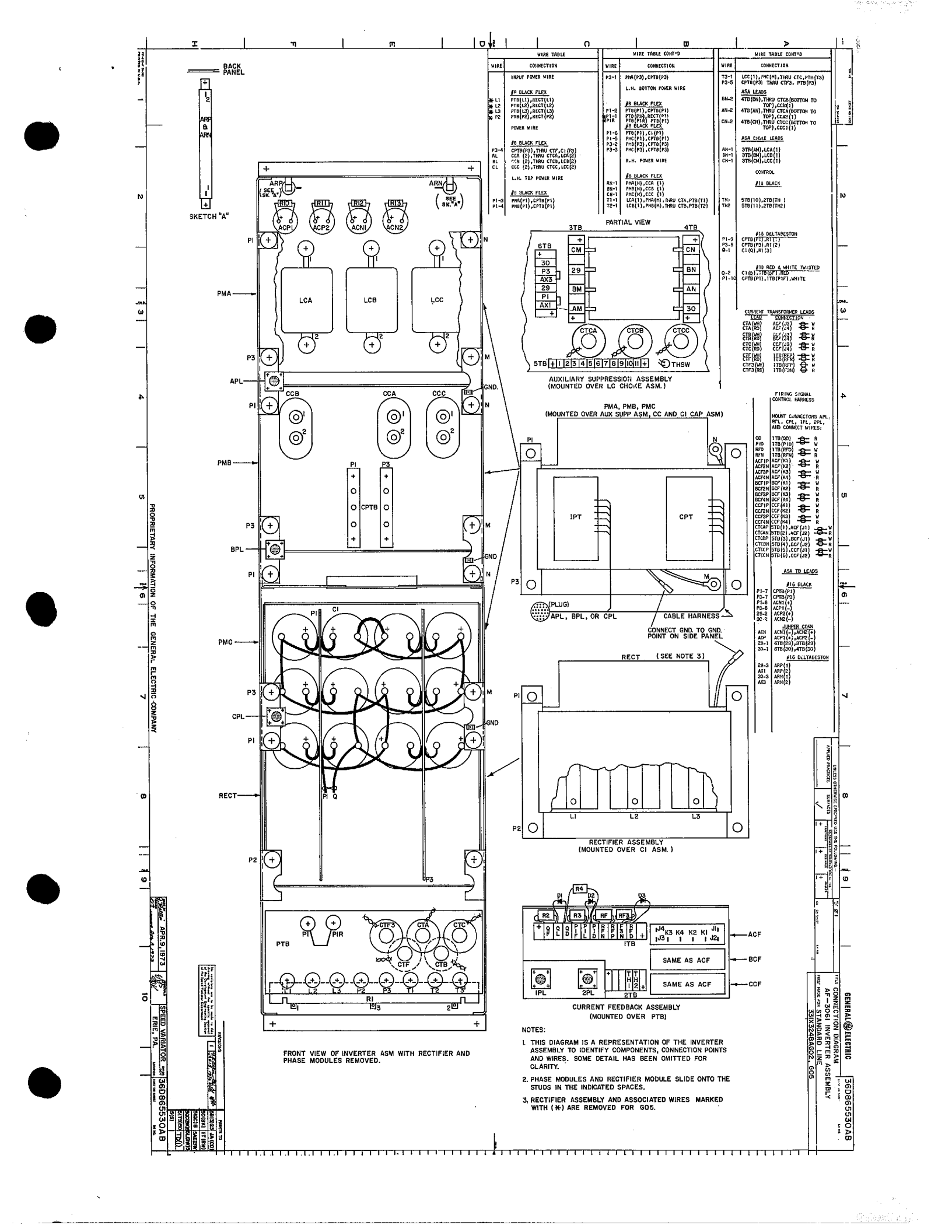
CONNECTION DIAGRAM

AF-3061 INVERTER ASSEMBLY

FOR STANDARD LINE

33KX324BAG01,B G04

36D865531AA



NOTES:

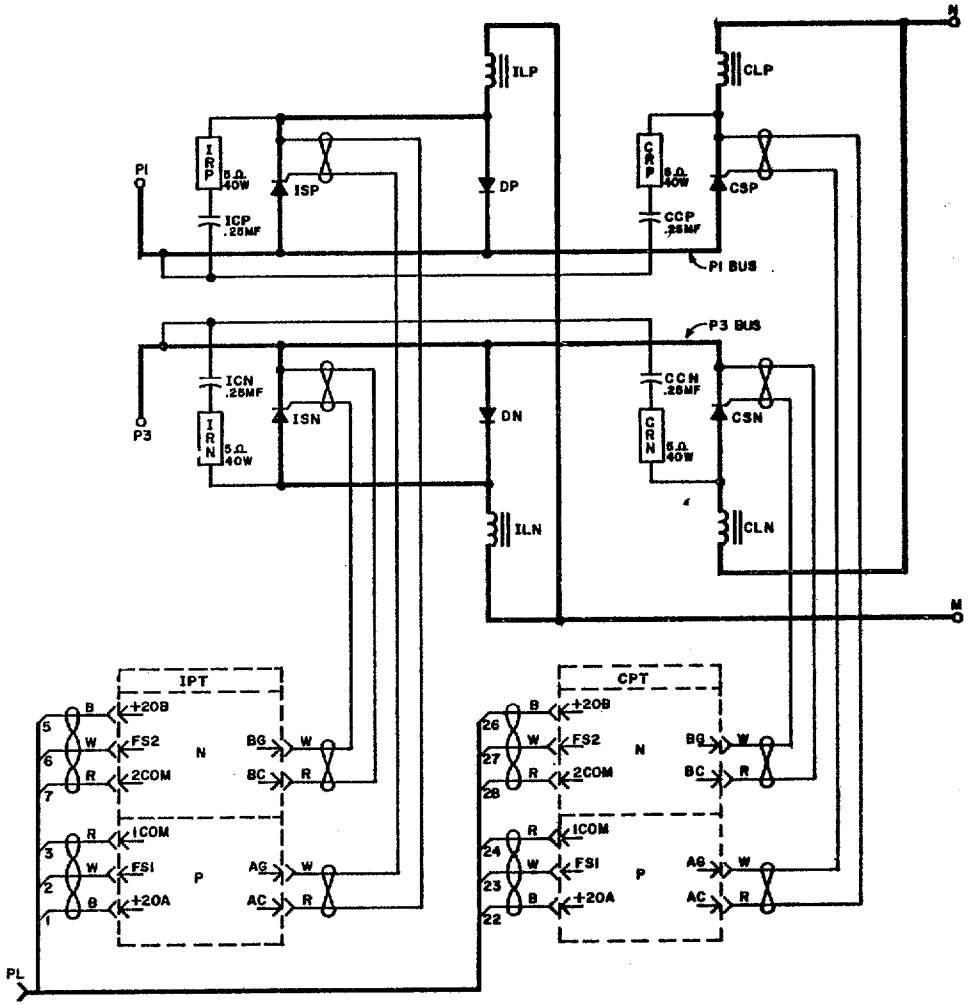
1. THIS DIAGRAM IS A REPRESENTATION OF THE INVERTER ASSEMBLY TO IDENTIFY COMPONENTS, CONNECTION POINTS AND WIRES. SOME DETAIL HAS BEEN OMITTED FOR CLARITY.
2. PHASE MODULES AND RECTIFIER MODULE SLIDE ONTO THE STUDS IN THE INDICATED SPACES.
3. RECTIFIER ASSEMBLY AND ASSOCIATED WIRES MARKED WITH (*) ARE REMOVED FOR G05.

GENERAL ELECTRIC

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES OR MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE ELEMENTARY DIAGRAM
AF-3061 PHASE MODULE
FIRST MADE FOR STANDARD LINE
331X323AAG01, G02, G03

CONT ON SHEET — SH NO.



NOTES

1. THIS DIAGRAM IS FUNCTIONALLY NOMENCLATED WITH PREFIXES OF I OR C TO INDICATE INVERTER OR COMMUTATION SECTION. SUFFIXES OF P OR N INDICATE POSITIVE OR NEGATIVE. THE BASIC NOMENCLATURE IS:
   - C — CAPACITOR
   - D — DIODE
   - L — CHOKE
   - PT — PULSE TRANSFORMER PRINTED CIRCUIT CARD
   - PL — CONNECTOR PLUG
   - R — RESISTOR
   - S — SCR
2. PULSE TRANSFORMER CARD CIRCUITRY IS NOT SHOWN FOR SIMPLIFICATION. FOR DETAILS REFER TO THE INSTRUCTION BOOK.
3. ELEMENTARY WIRING SYMBOLS.
   - R, W, B — INDICATES TWISTED WIRES, COLORS INDICATED BY LETTER.
   - 1, 2, 3 — INDICATES CABLE ASSEMBLY, NUMBERS ARE POINTS IN CONNECTOR PLUG PL.
   - →)(← — QUICK DISCONNECT "STAB-ON" CONNECTION.
4. REFER TO THE CONNECTION DIAGRAM FOR A PHYSICAL REPRESENTION OF THIS ASSEMBLY.
5. THE ASSEMBLY CAT. NO. GROUP DISTINCTION IS FOR MAXIMUM RATING:
   - G01 — 10 KVA
   - G02 — 25 KVA
   - G03 — 50 KVA

FOLD

DRAWING NO. 36C764814AA — CONT ON SHEET — SH NO.

REV NO. 0

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

PRINTS TO

58(B)23	JA(CD)
50(BK)	
5(QC)B	
5QCBK05	
5(TS)50	
5(S)	

DRAWN — DATE OCT. 9, 1972 — APPROVALS — SPEED VARIATOR ERIE, PA. — DEPT OR LOCATION — 36C764814AA — CONT ON SHEET — SH NO.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

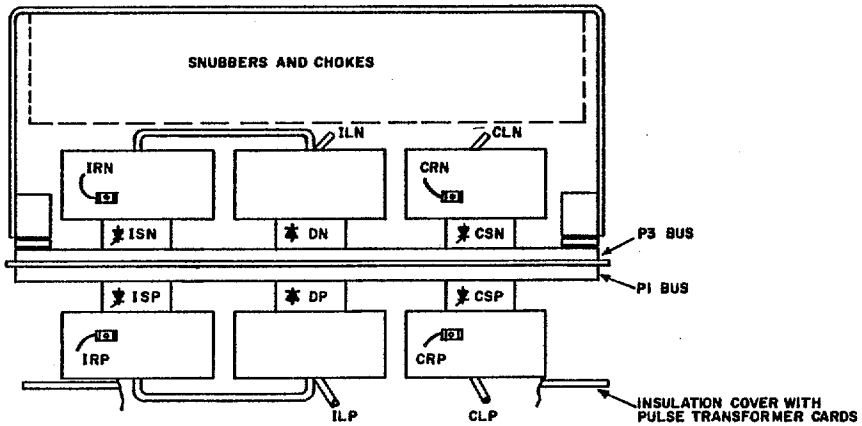
GENERAL ELECTRIC

36C 764 815 AA
CONT ON SHEET SH NO.

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

TITLE CONNECTION DIAGRAM
AF-306I PHASE MODULE
FIRST MADE FOR STANDARD LINE
331X323AAG02, G03

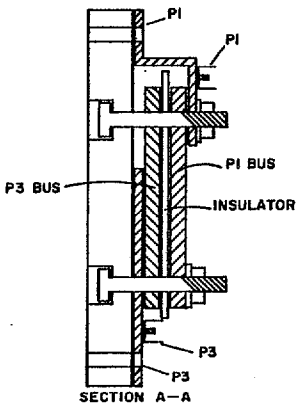


TOP VIEW OF PHASE MODULE

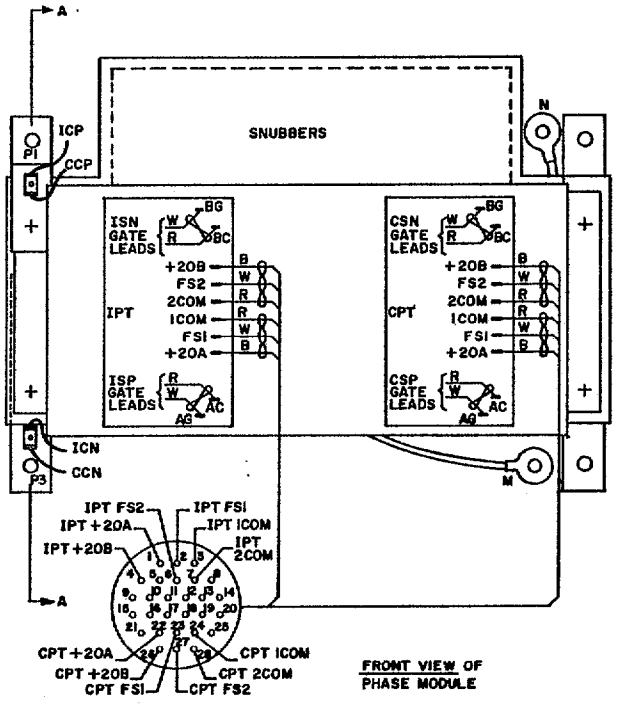
NOTES:

1. THIS DIAGRAM IS A REPRESENTATION OF THE PHASE MODULE TO IDENTIFY COMPONENTS, CONNECTION POINTS AND WIRES. SOME DETAIL HAS BEEN REMOVED FOR CLARITY.
2. GATE LEAD AND THE PULSE TRANSFORMER INPUT HARNESS ARE TWISTED WIRES INDICATED BY THE SYMBOL : W R B

   WIRE COLOR IS INDICATED FOR EASY IDENTIFICATION. GATE LEADS ARE WHITE, CATHODE LEADS ARE RED.

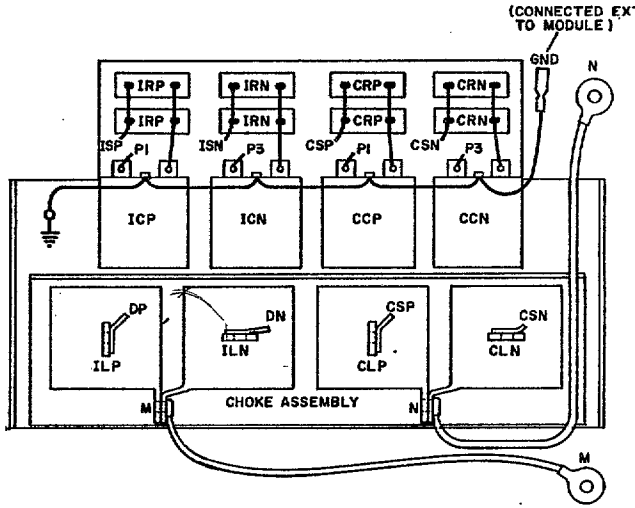


VIEW OF BUS ASSEMBLY



FRONT VIEW OF PHASE MODULE

PL
(VIEWED FROM WIRE SIDE)



FRONT VIEW OF SNUBBERS, CHOKES WITH HEATSINK ASSEMBLY REMOVED

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

PRINTS TO

5B(3)23	JA(CD)
5D(BK)	
5(QC)B	
5QCBK)5	
5(TS)50	
5(S)	

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

MADE BY G. Nesta OCT. 16, 1972
ISSUED G. DeLaguna Jr. 11-24-1972
APPROVALS
SPEED VARIATOR DIV. OR DEPT
ERIE, PA LOCATION
36C 764 815 AA
CONT ON SHEET SH NO.

GENERAL ELECTRIC

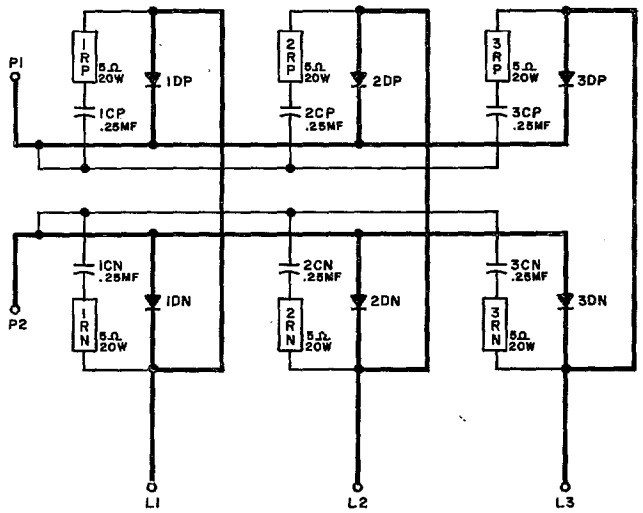
36C764812AA
CONT ON SHEET SH NO.

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

TITLE ELEMENTARY DIAGRAM
AF-3061 RECTIFIER MODULE

FIRST MADE FOR STANDARD LINE
331X322AAG02, G03



NOTES

1. THIS DIAGRAM IS FUNCTIONALLY NOMENCLATED WITH PREFIXES INDICATING PHASE (1, 2, OR 3) AND SUFFIXES OF P OR N INDICATING POSITIVE OR NEGATIVE. THE BASIC NOMENCLATURE IS:
   C – CAPACITOR
   D – DIODE
   R – RESISTOR
2. REFER TO THE CONNECTION DIAGRAM FOR A PHYSICAL REPRESENTATION OF THIS ASSEMBLY.
3. THE ASSEMBLY CAT. NO. GROUP DISTINCTION IS FOR MAXIMUM RATINGS:
   602 50 KVA 230AC
   603 60 KVA 460AC

DRAWING NO. 36C764812AA
CONT ON SHEET SH NO.
REV NO. 0

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

PRINTS TO	
58(3)23	JA(CD)
5D(BK)	
5(QC) B	
5QCBK5	
5(TS)50	
5(S)	

MADE BY G. Mattis SEPT 27, 1972
ISSUED

APPROVALS

SPEED VARIATOR DIV OR DEPT 36C764812AA

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

GENERAL ELECTRIC

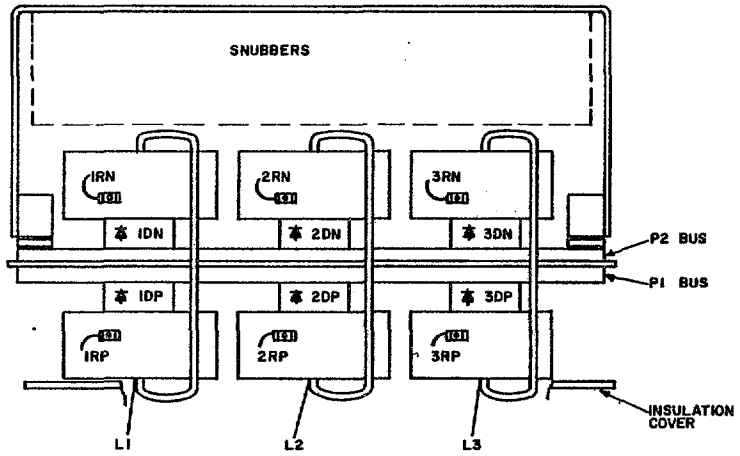
36C 764 813 AA
CONT ON SHEET SH NO.

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

TITLE CONNECTION DIAGRAM
AF-3061 RECTIFIER MODULE

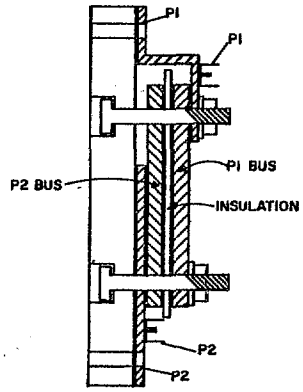
FIRST MADE FOR STANDARD LINE
331X322AAG02,003



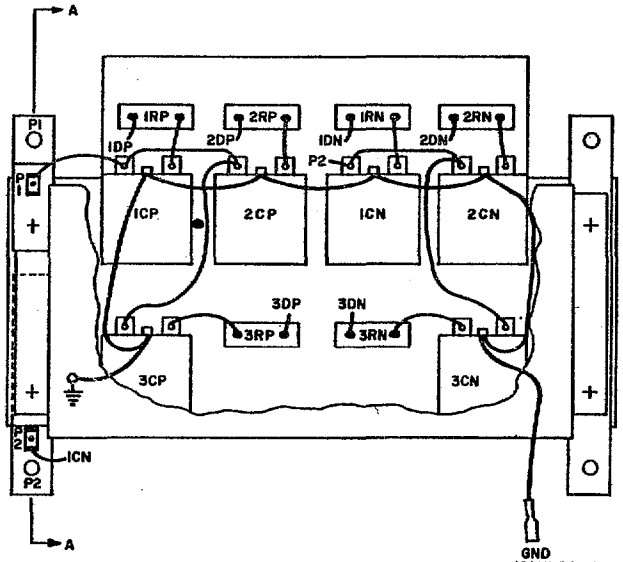
TOP VIEW OF RECTIFIER MODULE

NOTES:

1. THIS DIAGRAM IS A REPRESENTATION OF THE RECTIFIER MODULE TO IDENTIFY COMPONENTS, CONNECTION POINTS AND WIRES. SOME DETAIL HAS BEEN REMOVED FOR CLARITY.



SECTION A-A
VIEW OF BUS ASSEMBLY



FRONT VIEW OF RECTIFIER MODULE
WITH CUTAWAY SHOWING SNUBBERS

GND
(CONNECTED EXTERNAL TO MODULE)

FOLD

DRAWING NO. 36C 764 813 AA
CONT ON SHEET SH NO.
REV NC

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

1	R. Novak Feb. 24, 1975	EHS	GPS

PRINTS TO

5B(3)23	JA(CD)
5D(BK)	
5(QC)B	
5QCBK)5	
5(TS)50	
5(S)	

DRAWN BY	DATE	APPROVALS		DIV OR DEPT	DRAWING NO.	SH NO.
G. Martin	OCT. 10, 1972	EHS		SPEED VARIATOR	36C 764 813 AA	
ISSUED	J.D. Kamm Nov 24, 1972		KLP	ERIE, PA. LOCATION	CONT ON SHEET	

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

GENERAL ELECTRIC

36C764112AB

CONT ON SHEET | SH NO.

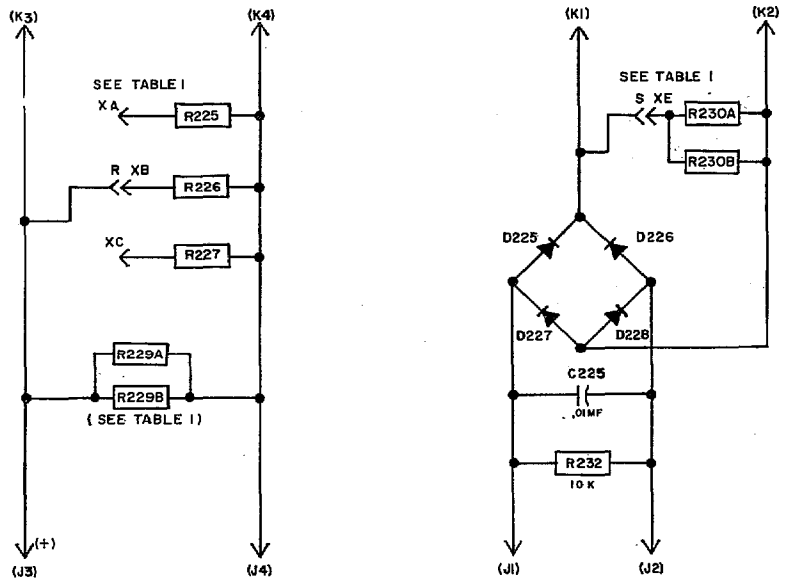
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
718A515	✓	±	±	±

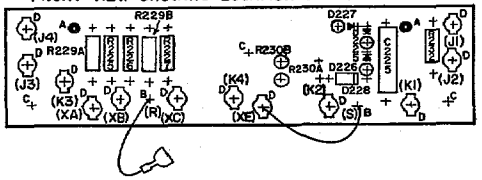
TITLE PRINTED CIRCUIT DIAGRAM
CURRENT FEEDBACK

FIRST MADE FOR STANDARD LINE
193X199BAG01, G02, G03

SCHEMATIC DIAGRAM



FRONT VIEW SHOWING LOCATION OF COMPONENTS



HOLE TABULATION

ALL HOLES .040 DIA. EXCEPT THE HOLES TABULATED BELOW

LOC.	DIA.	QUAN.
A	.157	2
B	.062	2
C	.187	3
D	.101	12

NOTE: 1. CARD SIZE, 4.900 $^{+.000}_{-.015}$ X 1.300 $^{+.000}_{-.015}$

NOTES

TABLE I

GROUP NUMBER	RESISTOR NOMENCLATURE AND VALUE						
	R225(XA)	R226(XB)	R227(XC)	R229A	R229B	R230A	R230B
G01	475Ω *	100Ω *	56.2Ω *	82.5Ω *	680Ω	121Ω *	100Ω *
G02	470Ω.	56.2Ω *	33.2Ω *	27.4Ω *	221Ω *	56.2Ω *	392Ω *
G03	10K	33.2Ω *	18.2Ω *	12.1Ω *	270Ω	27.4Ω *	560Ω

* — INDICATES 1% RESISTOR

TABLE 2

GROUP NUMBER	RESISTANCE BETWEEN K3 & K4 USING JUMPER R			RESISTANCE K1 TO K2
	XA	XB	XC	
G01	63.6Ω	42.4Ω	31.9Ω	54.8Ω
G02	23.2Ω	17.0Ω	14.1Ω	49.1Ω
G03	11.6Ω	8.6Ω	7.1Ω	26.1Ω

SCALE	ETCHED CIRCUIT BOARD DWG.	SPACE UNIT REQUIREMENT FRONT	BACK
FULL	942B519AB 004		

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

1 D. Mattia 9-13-72 ARB

REV NO. 1

PRINTS TO

5QC(BK)	5B(S)
5B(2)22	5D(BK)
5(S)	5P(W) 5Q
JA(CD)	5B(2)
1T(BW)	5(TS)50
5L(BW)5	5(QC)B

MADE BY D. Mattia SEPT. 8, 1972 APPROVED ARB

SPEED VARIATOR DIV OR DEPT

36C764112AB

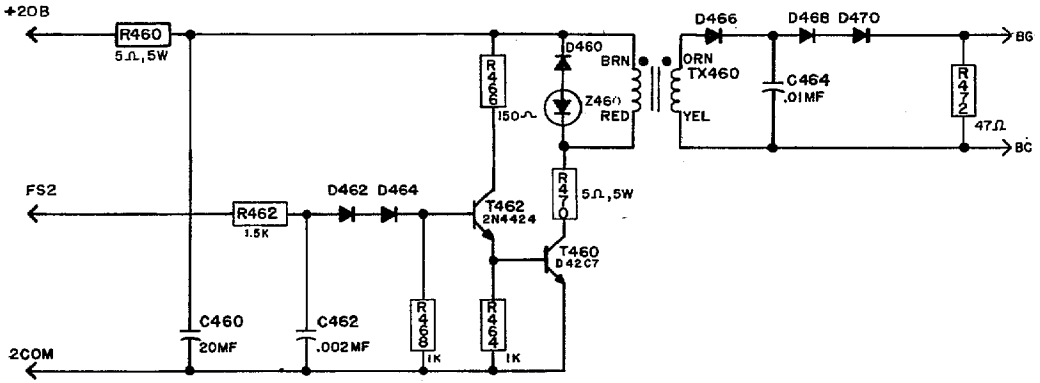
SCHEMATIC DIAGRAM

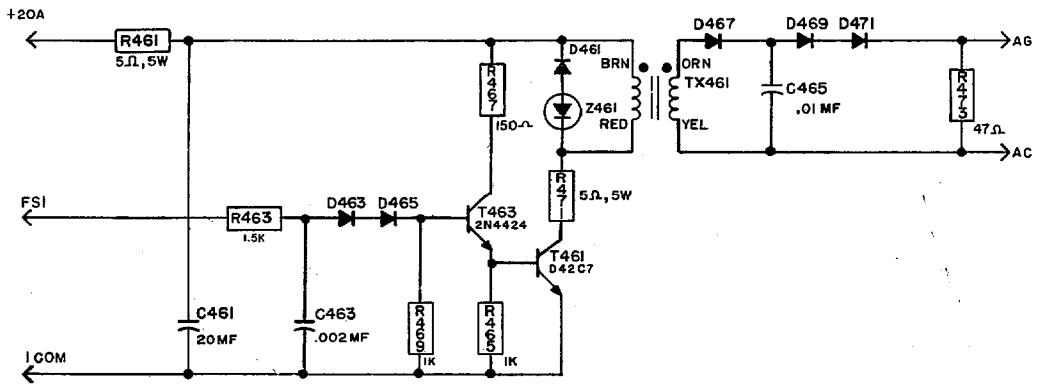
GENERAL ELECTRIC 36C762856AB

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING

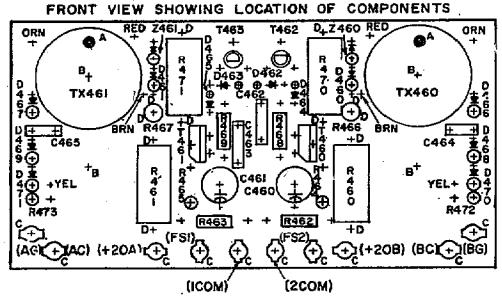
APPLIED PRACTICES	SURFACES	FRACTION	DECIMALS	ANGLES
718A515	✓	±	±	±

TITLE PRINTED CIRCUIT DIAGRAM
PULSE TRANSFORMER
MATL FOR STANDARD LINE
193X186ABG01





FRONT VIEW SHOWING LOCATION OF COMPONENTS



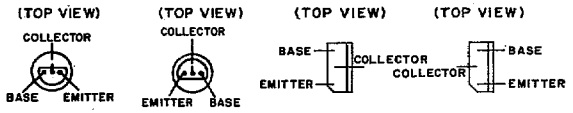
HOLE TABULATION

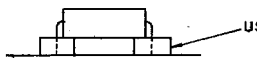
ALL HOLES .040 DIA. EXCEPT THE HOLES TABULATED BELOW

LOC.	DIA.	QUAN.
A	.157	2
B	.187	4
C	.101	10
D	.052	12

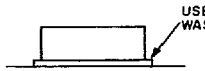
NOTE: 1. CARD SIZE $5.130^{+.000}_{-.015} \times 2.850 \pm .020$

TRANSISTOR LEAD SKETCHES





SKETCH FOR R460, R461, R470, R471



SKETCH FOR TX460, TX461

SCALE	ETCHED CIRCUIT BOARD DWG.	SPACE UNIT REQUIREMENT FRONT	BACK
FULL	942B521AA004	—	—

REVISIONS

No revisions are to be made to this drawing, without the specific approval of the Development Engineering Section of the Speed Variator Department.

5QC(BK)5	5 (S)
	5D(BK)
	5B(2)
	5(TS)50
	5(QC)B

36C762856AB
0

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

MFG'R'S APPROVALS
G. Novak Jan-25, 1972
D. Juneau JAN 26, 1972

APPROVALS
SPEED VARIATOR
ERIE, PA

36C762856AB

GENERAL ELECTRIC 36C762897B

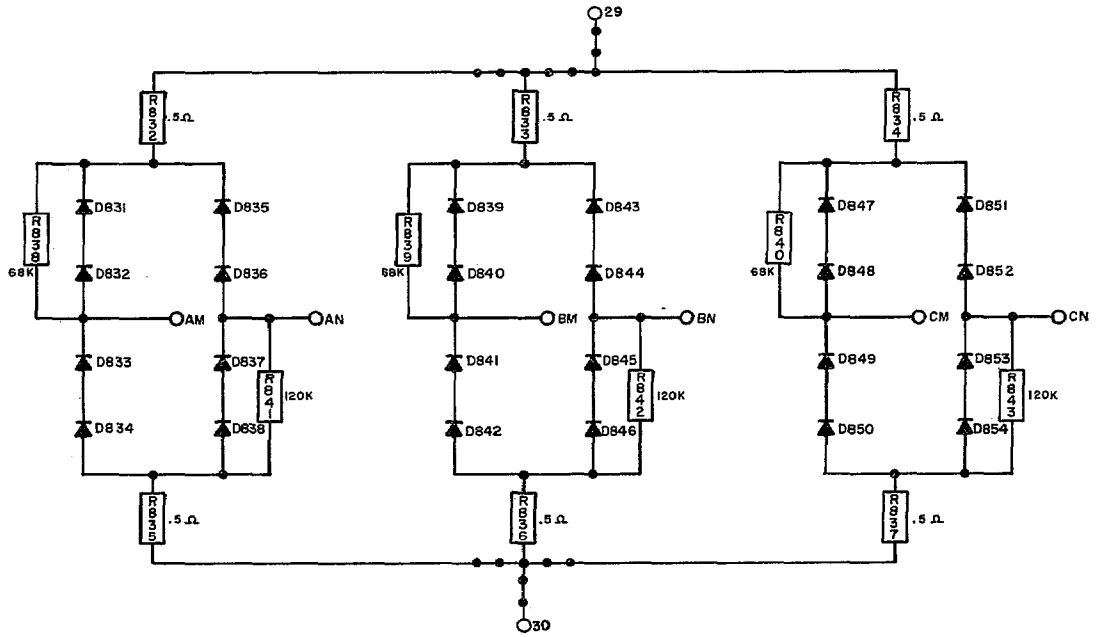
CONT ON SHEET SH NO.

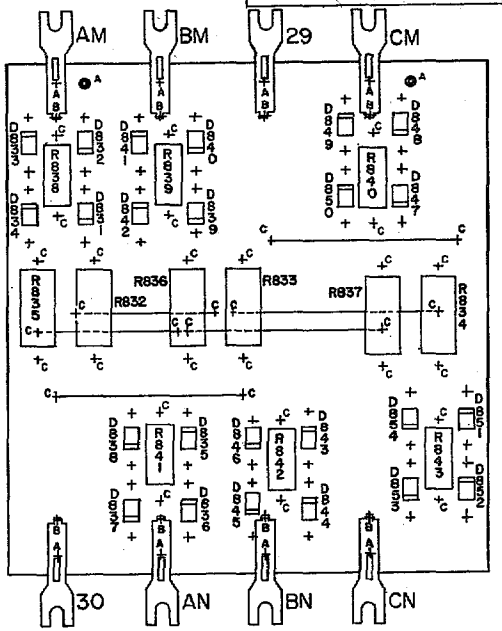
SCHEMATIC DIAGRAM

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:-				
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
718A515	✓	±	±	±

TITLE PRINTED CIRCUIT DIAGRAM
AUXILIARY SUPPRESSION

FIRST MADE FOR STANDARD LINE
193X1968AG01





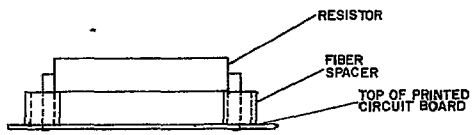
FRONT VIEW SHOWING LOCATION OF COMPONENTS

NOTE
1. CARD SIZE, $5.500^{+.000}_{-.015}$ X $5.130^{+.002}_{-.008}$

HOLE TABULATION

ALL HOLES .062 DIA. EXCEPT THE HOLES TABULATED BELOW

LOC.	DIA.	QUAN.
A	.157	10
B	.101	8
C	.052	36



MOUNTING SKETCH FOR RESISTORS R832, R833, R834, R835, R836, AND R837

CONT ON SHEET 36C762897BA SH NO.

REV NO. Ø 1

SCALE	ETCHED CIRCUIT BOARD DWG.	SPACE UNIT REQUIREMENT FRONT	BACK
FULL	9428595AA004	—	—

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

1	July 23 1973 G.P. Bahr		

PRINTS TO

5QC(BK)5 (S)
5D(BK)
5B(2)22
5(TS)50
5(QC)B

MADE BY G. Marsh FEB. 20, 1973
ISSUED G. Marsh July 27, 1973

APPROVALS

SPEED VARIATOR DIV OR DEPT
ERIE LOCATION

CONT ON SHEET 36C762897BA SH NO.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY